INDUS UNIVERSITY

Mid Semester Examination

Institute	IITE	Exam Month / Year	March 2026
Program	B.Tech.	**Branch**	CE/CSE/IT
Semester	4	**Regular / Repeater**	Regular
Course Code	CE0404	**Course Name**	Computer Organization & Architecture
Date	27/03/2026	**Time**	1:00 pm to 2:30 pm
Day	Friday	**Maximum Marks**	40

Instructions :

Attempt all questions.
Make suitable assumptions wherever necessary.
Figures to the right indicate full marks.

Q 1 A	List difference between Von Neumann and Harvard architectures in terms of performance and design.	04
Q 1 B	List different types of registers used in computer systems, describe any four.	04
	OR	
Q 1 B	In a particular system, suppose we can upgrade its disk drives to make them work 150% faster. Processes spend 70% of their time running in CPU and 30% of their time waiting for disk drive service. What would be the expected speedup according to Amdahl's law?	04
Q 2 A	Discuss the difference between RISC and CISC	04
Q 2 B	Draw the internal architecture of the 8085 microprocessor.	06
	OR	
Q 2 B	Design the Timing diagram for STA 526A. Opcode for (STA)= 32H.	06

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Q 2 C	List various types of addressing modes, discuss suitable examples.	06
	OR	
Q 2 C	Discuss different types of flag registers used in 8085 microprocessors.	06
Q 3 A	Explain Flynn's Classification of Computer Architecture.	04
Q 3 B	Explain 5-stage pipeline with diagram.	06
	OR	
Q 3 B	Explain arithmetic pipeline.	06
Q 3 C	List the Different types of pipeline hazards and explain them in detail.	06
	OR	
Q 3 C	Explain instruction pipeline.	06

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